Prior Authorization Intake Workflow Agent

FDE Cohort 5 Qualification Capstone — Track 2 · Vidhanshee Kakkar · August 2026 ·
SYNTHETIC DATA ONLY — NO PHI

Problem brief

Target user. The prior-authorization (PA) intake coordinator at UHC who receives PA requests from providers (portal, fax, phone) and must decide, case by case, whether a request is complete enough to enter clinical review.

Workflow pain point. Intake completeness is checked manually today, and incomplete requests are the single cheapest problem the PA chain fails to catch early. An incomplete packet doesn't cost one delay — it starts a provider round-trip cycle where each exchange adds days; it produces avoidable denials that are later overturned on appeal (pure rework on both sides); it burns the regulatory turnaround clocks CMS is tightening through 2026–27; and it feeds the provider abrasion UHC has publicly committed to reducing. Catching a missing TB screening at intake on day 0 is roughly an order of magnitude cheaper than expediting the same case after a denial on day 12.

Value hypothesis. An assistant that detects *everything* missing the moment a request arrives, drafts one consolidated provider follow-up (one round-trip instead of several), and routes the case correctly — with a human approving every outbound action — reduces round-trips per case and time-to-clinical-review without removing human accountability.

Success metrics. Missing-info detection precision/recall vs a golden set (recall ≥95%: a missed gap becomes a downstream delay; precision matters because false asks create new provider abrasion) · routing accuracy · follow-up letter rubric pass rate (covers every finding, invents nothing) · reviewer time per case.

Constraints & assumptions. Synthetic data only (no PHI); must run end-to-end for a reviewer with zero API keys; 6-CPT policy catalog standing in for the policy system of record; static eligibility snapshot standing in for the eligibility service; LLM spend capped by a hard budget guard.

What was built

> **fax text → extraction → completeness check → duplicate check → follow-up drafting → routing → HUMAN APPROVAL → audit log**
> Plus: event-driven intake (drop a file, it enters the queue), a review console (queue · dashboard · audit · evals tabs), and per-call LLM cost metering with a hard budget + runaway-loop cap.

The core design judgment — deterministic wherever a rule can decide; LLM only where language understanding is required. Field presence, NPI Luhn checksum, ICD-10/CPT format, eligibility lookup, duplicate detection, and all routing are plain auditable rules. The model does exactly two things: judging whether free-text clinical notes satisfy the procedure's policy requirements, and drafting the provider letter. Every model output is schema-constrained, verified by a code rubric (with one feedback retry, then a safe template fallback), and human-approved before any external effect. On any API failure, safety refusal, or budget breach the system degrades to fully offline deterministic mode — the workflow never blocks on a model.

Example that carries the value: a request for adalimumab (biologic) documents the RA diagnosis and the failed methotrexate trial, but never mentions the TB screening the policy requires. Keyword matching cannot reliably catch this; the model reads the notes and flags exactly that gap, and the drafted letter asks for exactly that — nothing invented, verified in code.

AI evidence

Model & tool choices

- Provider ladder: Anthropic Claude (`claude-opus-5`) → Google Gemini free tier (`gemini-3.5-flash-lite`) → offline deterministic fallbacks. One 30-line function per provider behind one interface.
- Structured outputs (JSON schema) on every call — no fragile text parsing.
- Prompts are defensive: "a requirement is met only if explicitly documented"; extraction "returns null for missing fields — never invents".
- Verification loop (loop-engineering "Loop 2"): draft → code rubric → one retry with the specific failures as feedback → template fallback.

Data approach

- Self-generated synthetic dataset: 27 packets + 2 fax documents across 15 scenario types (complete, missing/invalid fields, insufficient clinical notes per three different policies, expedited without justification, duplicates, fax channel). Seeded and reproducible.
- Golden labels emitted *by construction*, readable only by the eval harness — the pipeline cannot see its own answer key.

Human review checkpoints

- One approval gate at the sole point of external effect: a named reviewer approves / edits / rejects every proposal (review console or CLI).
- Append-only audit log of every proposal and decision (actor, timestamp).
- Reviewer letter-edits are captured — the labeled improvement signal for the next iteration.

Known failure modes & risks (measured, not hidden)

- Offline keyword fallback misses negation ("no PT attempted") — explicitly a fallback; LLM mode handles it.
- Rate limiting once silently diluted an "LLM-mode" eval with offline fallbacks — results looked perfect and weren't. Fixed with retry-waits *and* a call-integrity metric that proves which path served every call.
- A provider listing a model ≠ the model being servable (Gemini 404'd a listed model) — caught by the integrity metric; models are probed through the production code path before eval runs.
- Golden labels share provenance with the generator; mitigated by hand-reviewed scenarios and independent unit tests.

Measured results (29-case golden set)

Mode	Missing-info P / R / F1	Routing	Letter rubric	LLM call integrity
Offline (deterministic)	100% / 100% / 100%	29/29	16/16	n/a
LLM (Gemini, 26-case run)	100% / 100% / 100%	26/26	15/15	**38/38 served by LLM**

Offline 100% is a consistency check of the deterministic path against by-construction labels. The LLM run's value shows in *what* it catches (semantic gaps like the undocumented TB screening) — and in the integrity column, which exists because an earlier run was silently diluted. 26 automated tests cover the validators, router priorities, extraction, duplicate logic, and budget guard.

Enterprise readiness

Dimension	Prototype	Production path
Data classification	Synthetic only, zero PHI anywhere in the repo	Intake packets are PHI/HIPAA: BAA with the model provider, zero-retention API terms, encryption in transit and at rest
Access control	Local run; keys in gitignored `.env`; runs with no keys at all	**SSO + RBAC, least privilege across four roles** — intake coordinator (approve/edit only), clinical reviewer (own specialty queue), platform engineer (deploy, no approval rights: separation of duty), compliance (read-only). Approvals record the authenticated identity, so they are attributable. Secrets in a manager, injected at runtime, never logged. The agent holds **no write access to any system of record**; its only outputs are a proposal and, post-approval, a letter dispatched by the outreach service
Audit & logging	Append-only JSONL of every proposal and human decision	Ship to the enterprise log store with immutable retention
AI controls	Schema-constrained outputs · code-rubric verification · human gate before any external effect · deterministic routing. **Containment:** the model can only emit extracted fields, unmet-requirement IDs, or letter text — it cannot alter a route, approve a case, or send anything, so an injected instruction in clinical notes has no reachable effect	Same controls; add periodic sampled QA of approved letters and explicit prompt-injection testing
Cost controls	Every call metered (tokens + list-price cost) to a ledger; hard budget (`PA_BUDGET_USD`) + runaway-loop call cap degrade the agent to offline rules — a stuck loop cannot overspend	Shared metering service with per-team budgets and alerting
Handoff owner	PA intake operations (product owner) + platform engineering (service owner). Handoff package: DESIGN.md, BUILD_LOG.md (every decision and failure recorded live), this brief, and the eval harness as the regression gate.	

Cost & scaling sketch

The pipeline is stateless per case — horizontal scaling is trivial. LLM cost ≈ $0.02–0.04 per incomplete case at Opus list price (a fraction of a cent on Gemini); complete cases make one model call. Overnight backlogs fit the Batches API at 50% price. Against each avoided provider round-trip (days of turnaround, appeal rework), per-case model cost is noise.

Next iteration path

- **Image OCR in front of extraction** — the extraction interface already treats the fax as untrusted text, so a scan→text model (or multimodal Gemini reading the fax image directly) slots in without pipeline changes.
- **Policy catalog at scale** — load the hundreds-of-CPTs catalog from the policy system of record in place of the 6-CPT file.
- **Learn from the gate (loop-engineering Loop 4)** — reviewer edits and rejections in the audit log are labeled signal; feed monthly samples into prompt and policy updates with the eval harness as the regression gate.
- **Named scope cuts** — gold-carding trusted providers, richer duplicate similarity (beyond member+CPT window), letter-tone LLM-judge QA.

Repository: README for run instructions (zero keys needed) · demo recording accompanies this brief · UHC Tech / Optum — FDE Cohort 5 Capstone, Track 2.